

Major Project: Task Manager (Full-Stack MERN Application)

Assigned By: Alok Maddheshiya (MERN Program Mentor)

Objective

Develop a **Task Manager Web Application** using the **MERN Stack (MongoDB, Express.js, React.js, Node.js)** to demonstrate your understanding of full-stack development, REST API design, database modeling, and secure application architecture.

This is a **major project** aimed at evaluating your ability to build a complete production-ready application with **Authentication and Authorization**.

Project Overview

The Task Manager is a full-featured productivity web application where users can manage their personal tasks efficiently. Each user will have a **secure, private workspace** where they can create, update, delete, and track tasks.

The project must include a complete **authentication system using JWT (JSON Web Tokens)** and proper **authorization**, ensuring users can only access their own data.

Users should be able to organize tasks with attributes like **title, description, priority, due date, and completion status**, along with advanced features like filtering, searching, sorting, and task statistics.

Project Requirements

1. Authentication & Authorization

- Implement **User Registration (Sign Up)**
 - Fields: Name, Email, Password
 - Email must be unique
 - Password must be securely hashed using **bcrypt**
 - Return JWT token after successful registration
- Implement **User Login (Sign In)**
 - Validate credentials
 - Return JWT token on success

- Handle invalid credentials properly
 - Implement **User Logout**
 - Remove token from client side
 - Implement **Protected Routes**
 - All task APIs must require authentication
 - Use JWT middleware to verify user
 - Token must be sent in `Authorization: Bearer <token>`
 - **Authorization Rules**
 - Users can only access their own tasks
 - Prevent access/modification of other users' data
 - Validate ownership before update/delete operations
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2. Task Management (CRUD Operations)

- Create Task with:
 - Title (required)
 - Description (optional)
 - Priority (Low / Medium / High)
 - Due Date
 - Completion Status
 - Read Tasks:
 - Fetch all tasks of logged-in user
 - Fetch single task by ID
 - Update Task:
 - Edit task details
 - Only owner can update
 - Delete Task:
 - Delete single task
 - Delete all completed tasks
 - Toggle Completion:
 - Mark tasks as completed or pending
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3. Filtering, Searching, Sorting

- Filter by: All / Pending / Completed
 - Search tasks by title/description
 - Sort tasks by:
 - Newest / Oldest
 - Priority
 - Due Date
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4. Task Statistics Dashboard

Display real-time statistics:

- Total Tasks
 - Completed Tasks
 - Pending Tasks
 - Overdue Tasks
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5. Frontend Requirements

- Build UI using **React.js**
 - Responsive and modern design
 - Form validation with error messages
 - Toast notifications for actions
 - Task cards with priority indicators
 - Edit task using modal or separate page
 - Loading states during API calls
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6. Backend Requirements

- Build REST APIs using **Node.js + Express.js**
 - Use **MongoDB with Mongoose** for database
 - Implement:
 - JWT Authentication Middleware
 - Validation Middleware
 - Error Handling Middleware
 - Maintain clean folder structure and scalable architecture
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Functional Expectations

- Secure authentication system using JWT
 - Proper authorization and data protection
 - Fully working CRUD operations
 - Clean and user-friendly UI
 - Real-time filtering and searching
 - Well-structured backend APIs
 - Use of environment variables for sensitive data
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Submission Requirements

- Upload full project (Frontend + Backend) to **GitHub (public repo)**
 - Include a **README.md** with:
 - Project description
 - Tech stack
 - Setup instructions
 - API documentation
 - Host the project:
 - Frontend: Vercel / Netlify
 - Backend: Render / Railway
 - Database: MongoDB Atlas
 - Submit:
 - GitHub Repository Link
 - Live Hosted Project URL
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Submission Process

All project submissions will be accepted through the Google Form only.

Ensure that all required details are filled correctly before submitting.

Submission Form Link: <https://forms.gle/sbdB1u1z1vburoQL8>

Submission Deadline: 30th April, 2026

Important Note

This is a **Major Project**, and students must be able to explain:

- Authentication & Authorization flow
- Backend API design
- Database schema
- Frontend state management

Plagiarism is strictly not allowed. Students should write and understand their own code.

If you have any questions or require assistance, please feel free to reach out during session hours or contact the assigned mentor at alokmaddheshiya0010@gmail.com.